

Music Effects on Mental Health: Exploring the Impact of Music Metrics on Mental Health Issues

| Trinity College Dublin
| Data Management and Visualisations
| MSc Business Analytics



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1. Introduction

The connection between mental health and human emotions is a subject that has been increasingly explored as patterns emerge regarding music tastes and people's resultant psychological well-being.

As subscription services such as Spotify, Amazon Music and Deezer continue to dominate the music streaming space, these platforms provide insight into people's listening habits. We aim to analyse whether meaningful correlations can be found between these music streaming preferences and changes in mental health status to uncover whether music can serve as a positive method for emotional well-being.

Our primary question explores: *How does the age, music listening frequency and preferred music genres of people across the globe with specific mental health issues relate to music preferences and mood improvement?*

The following paper presents our methodology, data visualisations and entity relationship diagram to fully understand whether a causal connection can be derived. We can then leverage this information for future guidance for these streaming services, regarding what attracts people to their chosen platform. In particular, our insights will assist with what genres are especially important in improving the wellbeing of specific age groups.

2. Dataset Selection and Rationale

The datasets were all obtained from two Kaggle sources named *Music & Mental Health Survey Results* and *Global Music Streaming Trends & Listener Insights*.

The streaming dataset covers listening behaviors and preferences from 2018–2024 across major platforms (Spotify, Apple Music, Tidal). In terms of the survey, collected by questionnaires, the



Music & Mental Health Survey Results dataset includes the frequency of listening to 16 types of music, as well as four mental health scores for certain mental health issues (anxiety, depression, insomnia, and OCD).

The addition of the *Music & Mental Health Survey Results* dataset offers a multi-level view, allowing insights into how genre preference and time-of-day listening relate to mental-health outcomes. Both datasets were connected to explore the association between music listening behavior, streaming patterns, and mental health issues. Together, the combined dataset design brings multiple views together, helping us to understand and explain when, how, and why people listen to music, and how music influences mental well-being.

Figure 1 — Music & Mental Health — Data Overview Table

Age	Primary streaming service	Hours per day	Fav genre	Exploratory	BPM	Music effects	Instrumentalist	Composer
group people by age and see links with music habits or mood.	Respondents ' primary streaming platform used, comparing their cross platform behavior	The number of hours the respondent listens to music per day, which shows daily listening time and its relationship with emotions or study/work	Favorite or top music genre,which explain music taste and how different styles may influence emotions	If the respondents are actively explore new artists/genres? This may be related to emotions or creativity.	Preferred tempo (BPM).Used to study whether the beats of music affect emotions.	records how music helps them , allowing us to compare the impact of music on different mental issues.	If the respondent play an instrument regularly have an impact on feelings.	The respondent regularly composes music, which helps us explore how creative activity may relate to emotional expression.

Data Overview Table

Created using data provided by *Music & Mental Health Survey Results*.

Figure 2 — Global Music Streaming — Data Overview Table

User_ID	Age	Country	Streaming Platform	Top Genre
Unique identifier for each listener,allow us to link the same person across tables.	Age of the listener,which allowing us grouped by age and compare habits or preferences across ages.	Country where the user resides,enables regional comparisons and showing country-level differences.	The platform used,which helps compare behavior across platforms	The genre the listener streams the most.Shows taste patterns
Minutes Streamed Per Day	Number of Songs Liked	Most Played Artist	Subscription Type	Listening Time (Morning/Afternoon/Night)
Average daily listening time	Total liked songs by the user,make genre groups.	Artist the listener plays the most.Identifies different preferences and distinguishes niche vs. mainstream tastes.	Free or Premium subscription.Helps compare listening habits between free and paid users.	Time of day they listen the most . Helps us to understand daily routine patterns and find the association between specific time and emotions.

Data Overview Table

Created using data provided by *Global Music Streaming Trends & Listener Insights*.

3. Relationships Between Entities

The ERD shown in Figure 3 consists of four main entities, categorised by the user ID as their Primary Key (PK) and their corresponding attributes. The Logical Database Model (LDM) can be found in the Appendix as Figure 5.

Relationships Between Entities

- The **About User entity** encloses the main characteristics to describe the users of the music streaming platforms and their mental health condition, containing attributes: ID (PK), Age (FK), Favourite_genre (FK), Country, if they listen to music While_working, Music_effect on their mental health, if they have Anxiety, Depression, Insomnia, and OCD.

- The ***Music Type* entity** includes the types of music streamed, where the attributes are: ID (PK), Favourite_genre (FK), Instrumentalist, Composer, Exploratory, Foreign_language, and BPM.
- The ***Music Frequencies* entity** contains information about the frequency with which the user listens to a music genre. The attributes are: ID (PK), Streaming_platform (FK) used, Country, EDM, Folk, Gospel, Hip_hop, Jazz, K_pop, Latin, Lofi, Metal, Pop, R&B, Rap, Rock, and Video_game music.
- The ***Global Music Streaming* entity** encloses data about the listening habits of the users, and the attributes are: ID (PK), Age (FK), Streaming_platform (FK) used, Minutes_listened, Number_songs, Subscription, Listening_time during the day, Discovery_rate, and Repeat_rate of songs listened.

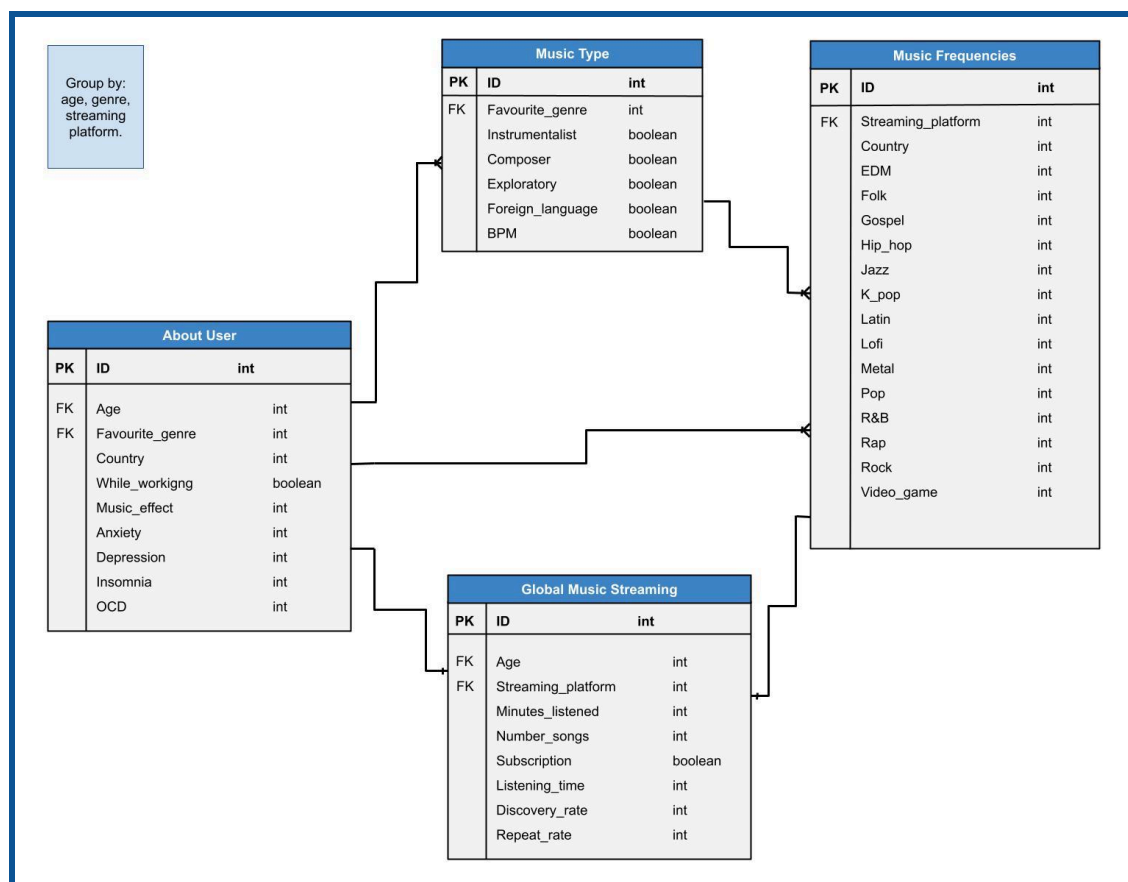
The *About User* entity is connected through a one-to-many relationship to the *Music Type* and to the *Music Frequencies* entities, as each user can have several music types and listen to music genres in different amounts. It is connected to the *Global Music Streaming* entity through a one-to-one relationship as each user is associated with a single set of streaming metrics (for instance, a user can only have one metric for minutes listened). *Music Type* is associated with *Music Frequencies* through a one-to-many relationship as a single music type can correspond to multiple genres listened to at varying frequencies. For example, an instrumentalist music type may include both Lofi and Jazz, which can each be listened to different numbers of times. Lastly, *Music Frequencies* is linked to *Global Music Streaming* through a one-to-one relationship, since the frequency of genre listening corresponds to a single streaming metric, similar to its connection with the *User About* entity.

Foreign Keys

The Foreign Keys in this model are *Favourite_genre*, *Age*, and *Streaming_platform*, and were chosen as these attributes connect the entities such that every user has a favourite music genre and listens to music from a defined streaming platform. The *Favourite_genre* attribute helps identify the types of music people listen to most and examine whether these preferences influence their mental health, while the *Streaming_platform* attribute provides insight into where

users access music and whether they hold subscriptions to those platforms. In addition, these variables are useful to see the frequency of which each genre is listened to, and explore other music metrics, such as the time of the day in which users listen to music the most, and for how long, among others. Finally, the *Age* attribute allows classification of users by age range based on their music habits and helps analyse whether certain age groups are more affected by mental health issues.

Figure 3 - Entity-Relationship Diagram

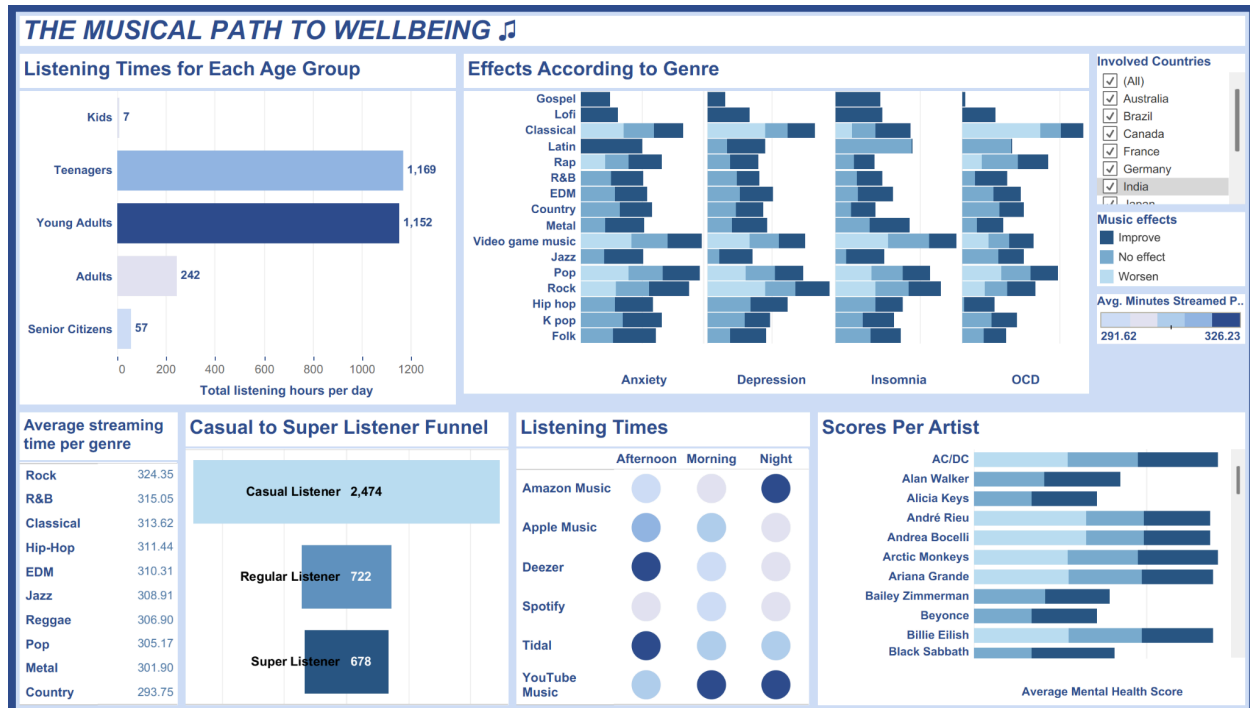


ERD

Created using data provided by Global Music Streaming Trends & Listener Insights on Google Drawings.

4. Visualisation and Graphic Ideas

Figure 4 - Tableau Dashboard



The overall intent of our dashboard in Figure 4, is to deliver the content in a clean yet effective way using simple visualisations with thoughtful additions. For the dashboard, we chose a card based layout over solid colour: the colours were chosen to create cohesion while also being colourblind friendly. Blue is often associated with calmness, trust and wellbeing, which aligns with our overall theme of mental wellbeing. The white spaces prevent cognitive overload and create clear visual grouping, which is also assisted by the margins created by the background which flows throughout the dashboard.

We chose the *Arial* as our primary font for the whole dashboard, as *Arial* is a Sans Serif font, meaning that it lacks decorative strokes in its design, thus providing high legibility, while also not being distracting. It is a widely used font which makes it compatible with multiple platforms. This, combined with the font being evenly sized and used properly in bold or italics, makes for a pleasant reading experience for the end user.



The dashboard is connected by the field *Age Group* in the *Listening Times for Each Age Group* through a common filter, such that when clicking on any of the age groups, the entire dashboard will respond with information pertaining to that particular age group. This is accompanied by a filter which can be used to look into the data with a specific country in mind. We have also added tooltips in all of the visualisations to either provide more data or context to the user in case the data cannot be explained just by using the axis.

These are the visualisations we have used:

1. Listening Times for Each Age Group (Bar Chart)

- a. Simple and direct comparison of listening hours among different age groups.
- b. Pie or line charts could distort magnitude differences; bars offer clear proportional representation.
- c. Shows which age groups engage the most.

2. Effects According to Genre (Grouped Bars)

- a. Allows side-by-side comparison of how different genres affect multiple mental health conditions (anxiety, depression, insomnia, OCD).
- b. Bar lengths more clearly communicate “direction and intensity” of effect.
- c. Demonstrates variation of emotional impact across genres.

3. Average Streaming Time per Genre (Horizontal Bar Chart)

- a. Easy to read genre names and compare across multiple categories.
- b. There’s no time sequence; this is categorical comparison data.
- c. Highlights which genres dominate in listening time (e.g., Rock and R&B).

4. Casual to Super Listener (Funnel)

- a. Perfectly visualises progression or engagement levels (Casual → Regular → Super).
- b. Funnel structure communicates conversion/transition logic better.
- c. Visualises listener intensity distribution effectively.

5. Data Results

Age

After grouping the sample by age, statistical results revealed that the teen and adolescent cohorts exhibited the longest music listening duration. At the same time, the prevalence of mental health issues also showed a decreasing trend with age maturity. This revealed a strong correlation between music listening duration across different age groups and mental health issues. This consistency in listening habits and mental health concerns across age cohorts served as the core link connecting the two datasets. Consequently, the multidimensional characteristics of younger groups regarding mental health issues and music listening habits represented the most valuable entry point for subsequent analysis.

Frequency

Additionally, individuals with higher mental health issue scores presented a greater tendency towards longer music listening sessions. We found that the surveyed sample group who indicated an improved result in ‘music effects’ also exhibited the highest average listening duration.

This further supports the hypothesis that an immersive music learning experience can enhance emotional regulation and even relieve stress. It also suggests that the intensity of listening habits, rather than mere frequency, may play a vital role in therapeutic effects.

Music Genres

Upon further analysing the therapeutic effects of different music genres, we uncovered the most valuable findings. Significant variations emerged for those who cited ‘improved’ effects across genres: Gospel and Lofi presented positive impacts on all four symptom categories, while Pop and Video game music yielded the most negative outcomes. Surprisingly, Classical music did not exhibit the generally assumed positive role in alleviating mental health issues, and in particular, Classical music showed a significant negative effect in OCD. Latin music also demonstrated



significant improvement in anxiety and depression symptoms, while showing no positive effects for those suffering with insomnia or OCD.

Focusing again on the younger age groups revealed further intriguing patterns. Classical music proved significantly more effective in alleviating mental health issues among young adults than among teenagers. Conversely, Hip-hop, Folk, and Jazz music demonstrated powerful therapeutic effects specifically for teenagers. Somewhat surprisingly, Gospel music received high praise from teenagers but found virtually no appeal among young adults.

Listener and Membership Type

From our funnel visualisation in Fig. 4., we can see that the difference between casual listeners and super listeners (those who listen for more than 3 hours per day) is minimal. In other words, many casual users already sit near the next tier in engagement. Therefore, casual users can be converted to super listeners easily through greater music preference targeting.

6. Our Recommendations and Conclusions

Our analysis of the datasets of the *Music & Mental Health Survey Results* and *Global Music Streaming Trends & Listener Insights* demonstrates that we can find specific correlations between music streaming habits and improved mental health benefits. Most significantly, listener age, listening frequency and music genre are useful indicators for streaming platforms seeking to target specific genres to different age groups.

Therefore, we can conclude that listening to music can have positive effects on various categories of mental health issues. However, different music genres exert significantly varying impacts on different symptom categories. Consequently, selecting specific music types will be the core factor in determining the success of music therapy promotion for individuals with diverse mental health concerns.

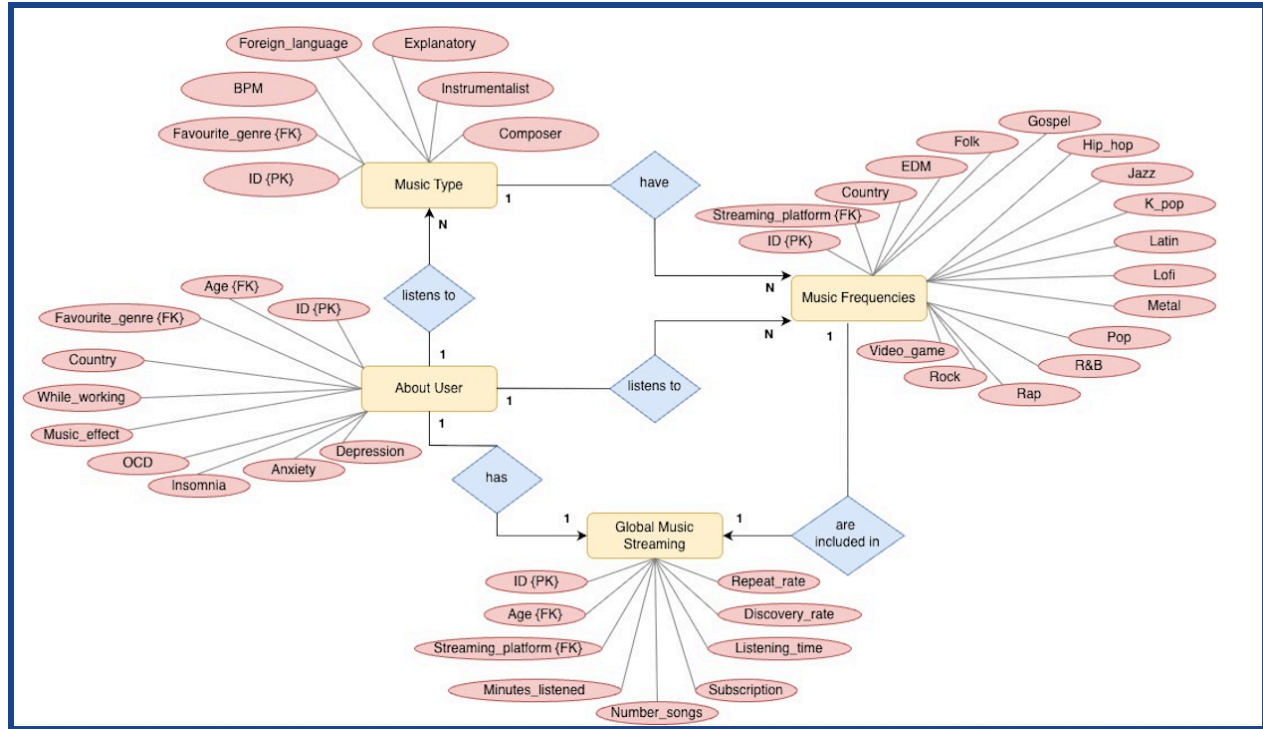


This means that listeners with mental health issues are a reachable target group for music streaming and market promotion, especially given that casual listeners can be easily converted to super listeners which will increase engagement with these subscription-based platforms. In particular, as our data shows, those in the younger age bracket are especially good targets for streaming platforms to promote specific genres of music which have an important positive impact on psychological wellbeing.



7. Appendix

Figure 5 - Logical Database Model



LDM

Created using data provided by Global Music Streaming Trends & Listener Insights on [Draw.io](https://draw.io).

Figure 6: Dashboard

Mental Health Score: $([OCD] + [Anxiety] + [Depression] + [Insomnia]) / 4$

Engagement Level:

```

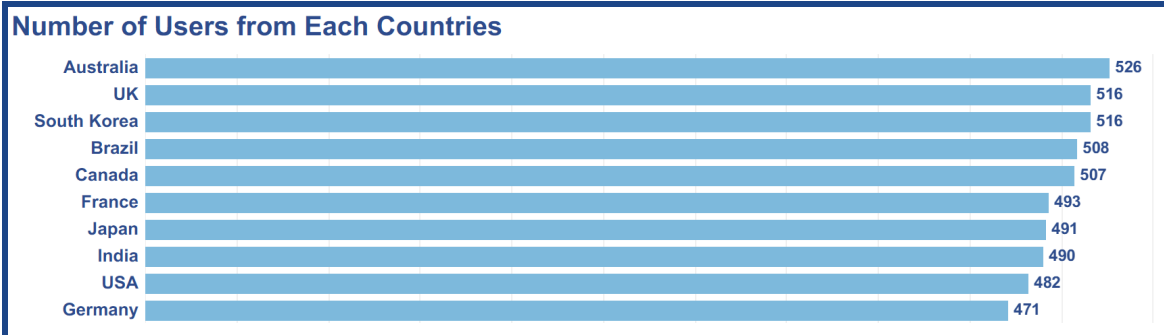
IF [Subscription Type] = "Free" THEN "Casual Listener"
ELSEIF [Subscription Type] = "Premium" AND [Minutes Streamed Per Day] <= 180 THEN
"Regular Listener"
ELSEIF [Subscription Type] = "Premium" AND [Minutes Streamed Per Day] > 180 AND
[Discover Weekly Engagement (%)1] > 60 THEN "Super Listener"
END
  
```

a) Table displaying the genre and average streaming time:

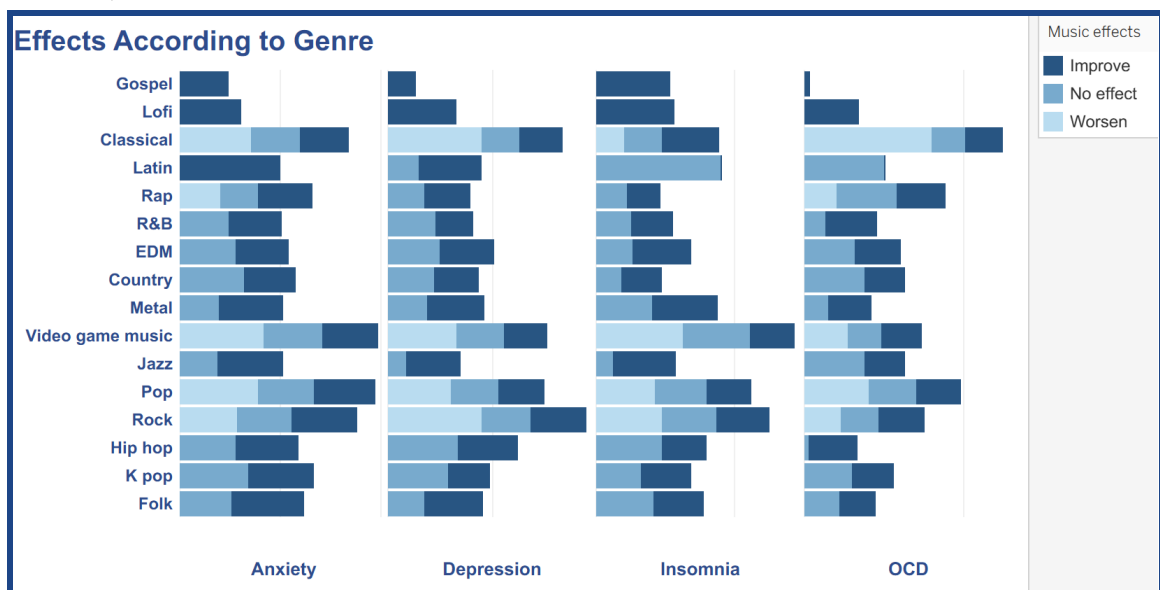
Average streaming time per genre

Rock	324.35
R&B	315.05
Classical	313.62
Hip-Hop	311.44
EDM	310.31
Jazz	308.91
Reggae	306.90
Pop	305.17
Metal	301.90
Country	293.75

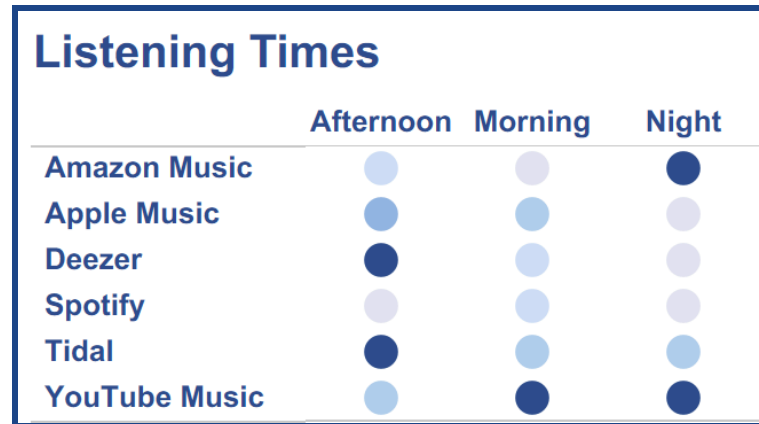
b) Horizontal bar graph displaying the number of users from each country:



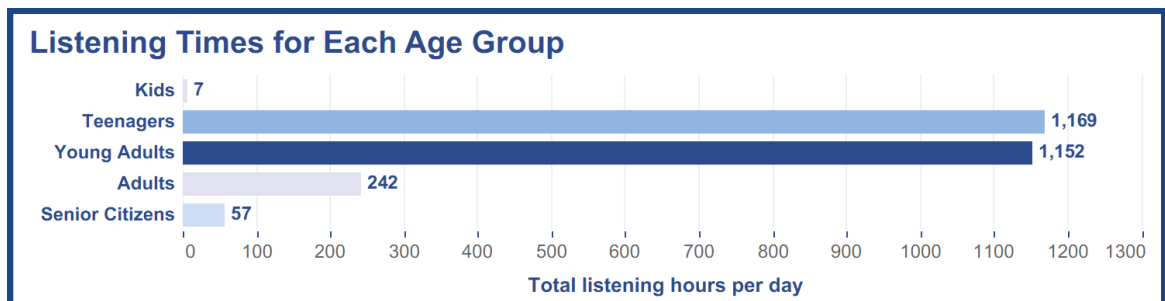
c) Effects of listening to each genre on 4 mental health variables (Anxiety, Depression, OCD, Insomnia):



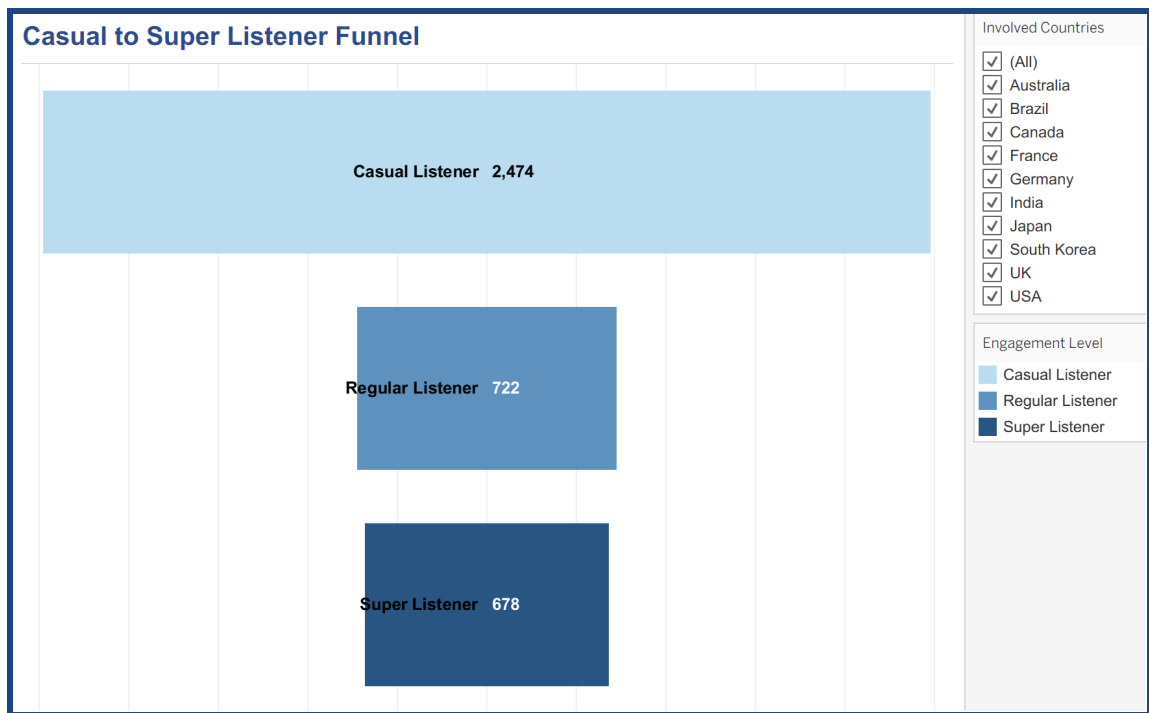
- d) Heatmap of listening times to see at what time of the day a platform is used the most by the users:



- e) Horizontal bar graph showing the amount of users from each age group:



- f) A funnel chart showing the transition from casual listeners to super listeners:



g) Horizontal bar graph showing the average mental health score by artist:

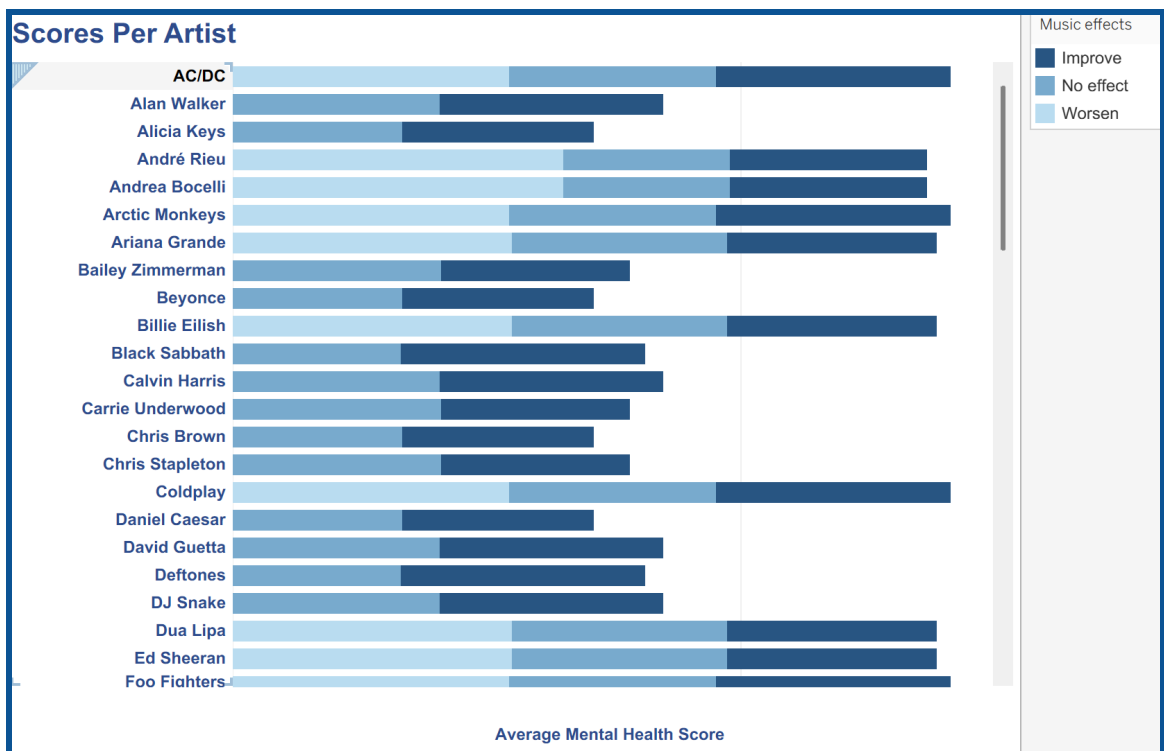
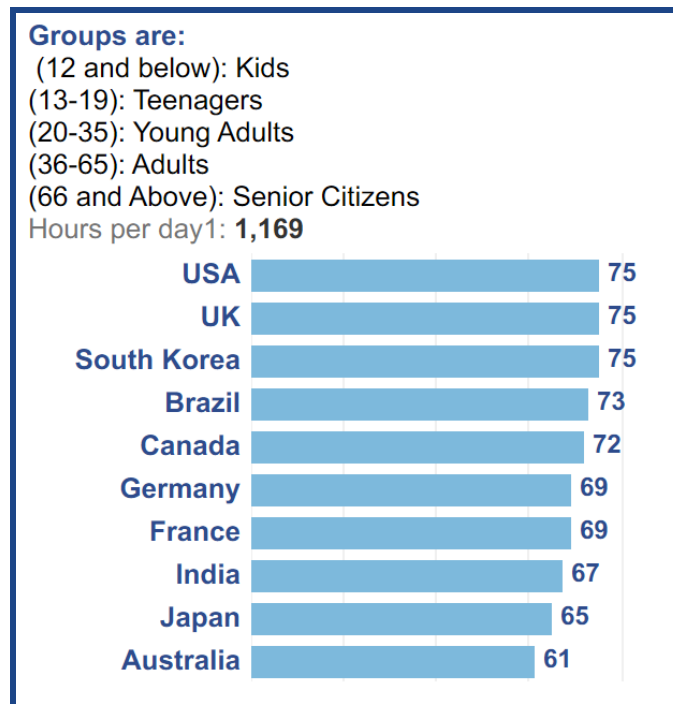


Figure 7 - Tooltips

Displays age-group definitions and country listening hours on hover:



8. Further Literature and Links

Rasgaitis, Catherine, 2022, 'Music and Mental Health Survey Results',
<https://www.kaggle.com/datasets/catherinerasgaitis/mxmh-survey-results>.

Soundankar, Atharva, 2024, 'Global Music Streaming Trends & Listener Insights',
<https://www.kaggle.com/datasets/atharvasoundankar/global-music-streaming-trends-and-listener-insights/data>

Link to Canva presentation,
https://www.canva.com/design/DAG2QJc80-4/HL2hwKC_d2fowH5OKz9EwQ/view?utm_content=DAG2QJc80-4&utm_campaign=designshare&utm_medium=link2&utm_source=uniquelinks&utlId=h1c20f03266

Link to Tableau Dashboard,
<https://public.tableau.com/app/profile/akshat.srivastava1989/viz/MentalHealthwtf/Dashboard1>